Critical error rate analysis for MSA-based variant phasing in nanopore amplicon sequencing

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Abstract

Amplicon consensus tools that detect intra-specimen sequence variants must distinguish real biological variation from sequencing error. We present a statistical framework for evaluating the significance of variants detected by MSA-based position phasing in Oxford Nanopore amplicon data. Rather than computing p-values against a fixed error rate, we derive the *critical error rate* p^*: the per-position sequencing error rate at which an observed variant pattern would become plausible under a null hypothesis of independent errors. Using conservative assumptions—total input reads as the sample denominator and Bonferroni correction over all amplicon positions—we show that variants with moderate read support require per-position error rates exceeding those of current nanopore chemistry to be explained as artifacts. We present results under both a conservative error model ($q = p$, maximally biased substitution) and a uniform model ($q = p/3$), and examine the effect of significance level on variant confidence across a range of operating conditions. We additionally provide minimum read-support thresholds at assumed error rates, enabling direct lookup of the evidence required for variant confidence.

1 Introduction

Oxford Nanopore sequencing of PCR amplicons is widely used for taxonomic identification, particularly in fungal DNA barcoding of the internal transcribed spacer (ITS) region [4]. After demultiplexing, reads from a single specimen are clustered and consensus sequences are generated to produce high-accuracy reference sequences suitable for database matching.

When a specimen contains multiple sequence variants—arising from heterozygosity, paralogous loci, mixed templates, or multi-copy rDNA polymorphism [1]—the clustering and consensus pipeline must resolve these into separate consensus sequences rather than collapsing them into a single ambiguous output. Speconsense [5] addresses this through MSA-based variant phasing: after initial clustering, it examines the partial order alignment [3, 7] within each cluster to identify positions where a substantial fraction of reads consistently differ from the consensus, then recursively splits the cluster into haplotype groups based on allele patterns at these positions.

A natural concern is whether detected variants represent real biological sequences or artifacts of sequencing error. If nanopore errors at a particular position happened to affect the same base in enough reads, the phasing algorithm could mistake this for a genuine variant. We address this concern by deriving the per-position error rate required to make each variant plausible as an artifact, and showing that this critical rate exceeds realistic nanopore error for variants with sufficient read support.

2 Methods

2.1 Null model

Under the null hypothesis, all reads in a cluster originate from a single true sequence, and observed base differences are independent sequencing errors. At each position, each read independently has probability p of containing an error.

We present results under two substitution models:

- **Conservative** ($q = p$): All errors at a position produce the same alternative base. This represents the worst case, accounting for nanopore substitution biases that favor certain error types [2, 10]. If errors are concentrated toward one alternative base, the probability of multiple reads independently producing the same incorrect base is maximized.
- **Uniform** ($q = p/3$): Errors are spread equally across the three alternative bases. This is more realistic for positions without strong substitution bias, and provides a less conservative bound.

2.2 Parameter definitions

- N: Total input read count for the specimen, before clustering. This is conservative because it represents the full pool from which a biased subsample could be drawn. In practice, speconsense uniformly presamples to $N \leq 1000$ reads, so this is bounded.
- M: Number of reads supporting the variant allele at a position. For $K > 1$, M is the number of reads supporting the variant allele at all K positions simultaneously (Section 2.5).
- L: Amplicon length in base pairs. For the ITS region, $L \approx 700$ bp. This determines the multiple testing correction.
- K: Number of variant positions distinguishing a given variant from all other variants produced from the same specimen. We use $K = 1$ (single-position variants) throughout as the conservative bound (Section 2.5).
- α: Significance level. We examine $\alpha = 0.05$ (per-variant), $\alpha = 10^{-3}$, and $\alpha = 10^{-5}$ (appropriate when correcting for thousands of variant calls across a sequencing run).

2.3 Critical error rate for single-position variants ($K = 1$)

For a single position where M of N reads share the same alternative base, the probability under the null model is given by the binomial survival function:

$$P(\text{position is artifact}) = P\big(\text{Binom}(N, q) \geq M\big) \tag{1}$$

where $q = p$ (conservative) or $q = p/3$ (uniform).

With Bonferroni correction over L independent positions, the probability that *any* position exhibits a variant-like pattern is bounded by:

$$P(\text{any artifact variant}) \leq L \cdot P\big(\text{Binom}(N, q) \geq M\big) \tag{2}$$

The critical error rate p^* is defined as the value of p satisfying:

$$L \cdot P\big(\text{Binom}(N, q^*) \geq M\big) = \alpha \tag{3}$$

where $q^* = p^*$ under the conservative model or $q^* = p^*/3$ under the uniform model. This is a single-variable equation solved numerically via root-finding on the log-scale survival function. If p^* substantially exceeds the actual per-position error rate of the sequencing platform, the variant cannot plausibly be explained by error.

2.4 Choice of significance level

A single variant call at $\alpha = 0.05$ has a 5% false positive probability under the null. In a typical sequencing run producing thousands of variant calls across hundreds of specimens, the run-level false positive rate can be substantial. For a run of S specimens producing an average of V variants each, the expected number of false positives is $\alpha \cdot S \cdot V$.

With $S = 1000$ and $V \approx 3$, $\alpha = 0.05$ yields ∼150 expected false positives per run. A Bonferroni correction to the run level suggests $\alpha \approx 0.05/3000 \approx 1.7 \times 10^{-5}$. We therefore present results at $\alpha = 10^{-5}$ as a practical run-corrected threshold alongside the per-variant $\alpha = 0.05$.

2.5 Multi-position variants ($K > 1$)

When a variant is distinguished from other variants at $K > 1$ positions, we define M as the number of reads that support the variant allele at all K positions simultaneously. Under the null hypothesis, these M reads must each independently produce the matching error at all K positions. With independent errors across positions, the probability that a single read matches the variant allele at all K positions is q^K, where q is the per-position error probability. The critical error rate equation becomes:

$$\binom{L}{K} \cdot P\big(\text{Binom}(N,\, (q^*)^K) \geq M\big) = \alpha \tag{4}$$

where $(q^*)^K$ is the joint per-read probability of matching the variant at all K positions, and the Bonferroni correction is over $\binom{L}{K}$ possible position combinations.

The effect is dramatic. At $N = 1000$, $M = 100$, $\alpha = 0.05$ (conservative model): $K = 1$ gives $p^* = 6.75\%$, while $K = 2$ gives $p^* = 24.19\%$—a 3.6× increase. The combinatorial correction grows from $L = 700$ to $\binom{700}{2} = 244{,}650$ (a 350× increase), but the joint error probability q^2 shrinks so rapidly that it overwhelmingly dominates. Under production settings ($p = 1.5\%$ assumed error, uniform model, $\alpha = 10^{-5}$, $N = 1000$), the minimum M for significance drops from 23 at $K = 1$ to 6 at $K = 2$ and 4 at $K = 3$.

The thresholds derived from this model are sensitive to the cross-position independence assumption. At the production operating point ($N = 1000$, $K = 2$, uniform model, $\alpha = 10^{-5}$, $p = 1.5\%$), the minimum $M = 6$ tolerates only a 2.2× increase in the joint error probability relative to the independent value q^2. Equivalently, if the conditional probability of error at one position given an error at another rises from the independent expectation of 0.5% to ∼1.1%, the $M = 6$ threshold loses significance. Nanopore errors exhibit sequence-context dependencies [2]: positions within the same k-mer context may share correlated error profiles, and a single problematic translocation event can affect multiple nearby bases. For variants whose distinguishing positions are distant in the amplicon (tens of bases or more apart), independence is reasonable and the joint model is well-supported. For positions clustered within a few bases, the effective exponent may be less than K, and the $K = 1$ analysis provides the safer bound.

The analysis that follows uses $K = 1$ throughout as the conservative bound. Multi-position variants are substantially more significant than this bound suggests.

3 Results

3.1 Critical error rates at fixed N

Figure 1 shows the critical error rate p^* as a function of variant read count M at $N = 1000$ (the standard presample size) for $K = 1$. Curves are shown for three significance levels under the conservative model ($q = p$), with the uniform model ($q = p/3$) overlaid at $\alpha = 0.05$ and $\alpha = 10^{-5}$ for comparison.

Under the conservative model at $\alpha = 0.05$, the critical error rate crosses the 5% reference line at approximately $M = 80$, and $M = 100$ yields $p^* = 6.8\%$. Even at the stringent run-corrected $\alpha = 10^{-5}$, $M = 100$ gives $p^* = 5.5\%$—still well above realistic nanopore error rates. The difference between significance levels is modest at high M, reflecting the exponential sensitivity of the binomial test.

Under the uniform model, the same cases yield $p^* = 20.3\%$ and 16.6% respectively, providing comfortable margins.

At low M, the test's power drops rapidly. The boundary where variants become questionable depends on N, but for typical coverage ($N = 100$–1000), variants with fewer than ∼20–30 supporting reads under the conservative model yield p^* values in the range of realistic nanopore error rates (1–5%), making them difficult to distinguish from artifacts.

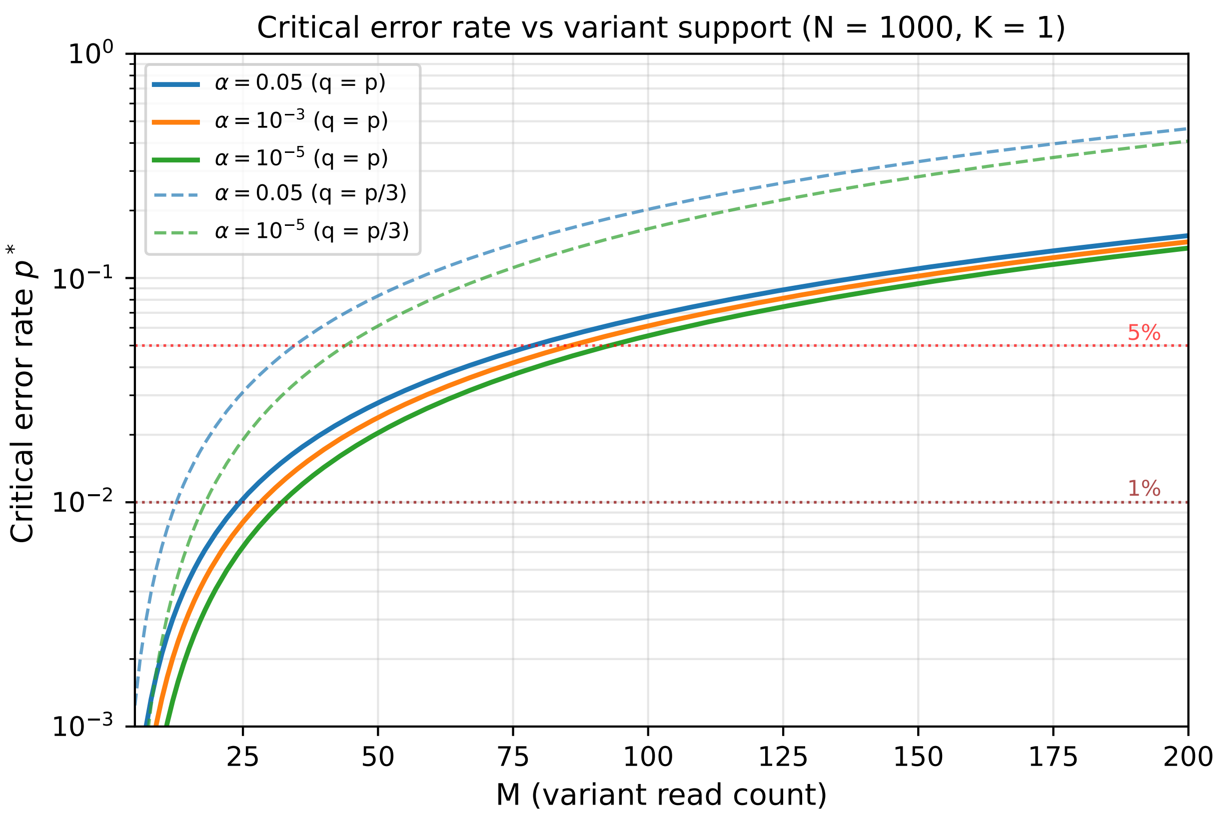

Figure 1: Critical error rate p^* vs. variant read count M at $N = 1000$, $K = 1$, $L = 700$. Solid lines: conservative model ($q = p$) at three significance levels. Dashed lines: uniform model ($q = p/3$) at $\alpha = 0.05$ and $\alpha = 10^{-5}$.

3.2 Effect of total read count

Figure 2 shows p^* as a function of M for several values of N under both error models ($\alpha = 0.05$). For a given M, larger N yields lower p^* because the same absolute count represents a smaller fraction of the total pool.

The side-by-side comparison illustrates the roughly threefold difference between error models. Under the uniform model, even relatively modest support ($M = 20$ at $N = 1000$) yields p^* above 1%. Under the conservative model, $M \approx 30$ is needed to reach the same threshold.

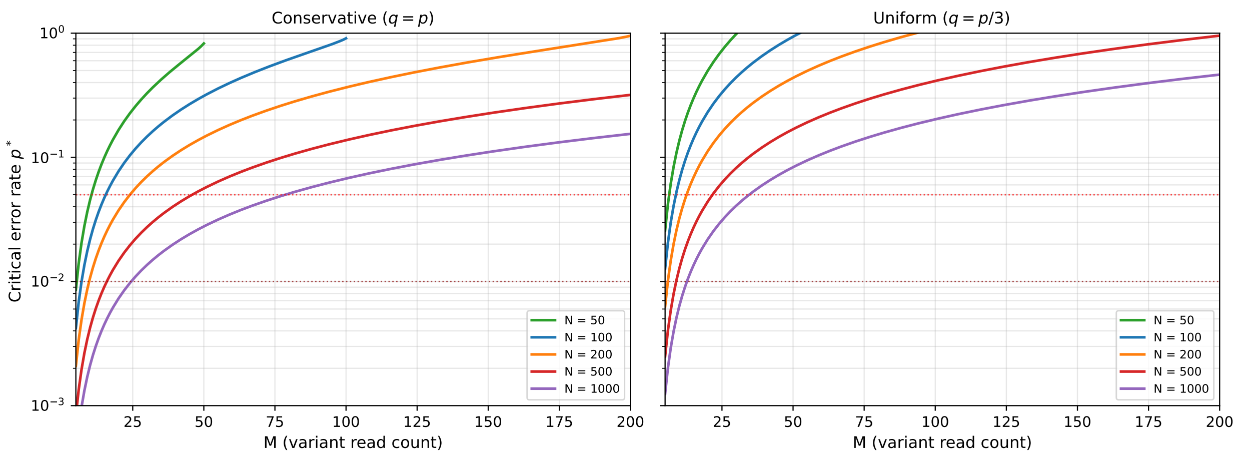

Figure 2: Critical error rate p^* vs. variant read count M for several total read counts N ($K = 1$, $L = 700$, $\alpha = 0.05$). Left: conservative model ($q = p$). Right: uniform model ($q = p/3$).

3.3 Summary of key values

Table 1 summarizes critical error rates for representative parameter combinations under both error models.

Table 1: Critical error rates p^* for representative parameter combinations ($K = 1$, $L = 700$). Values shown for both conservative ($q = p$) and uniform ($q = p/3$) error models.

			p^*		
N	M	α	$q = p$	$q = p/3$	Note
1000	100	0.05	6.75%	20.25%	Standard operating point
1000	100	10^{-5}	5.53%	16.59%	Run-corrected
100	10	0.05	2.18%	6.55%	Moderate coverage
100	10	10^{-5}	0.83%	2.50%	Run-corrected, moderate coverage
50	5	0.05	0.86%	2.57%	Boundary case
50	5	10^{-5}	0.15%	0.44%	Run-corrected boundary

3.4 Minimum read support at assumed error rates

While Table 1 shows the error rate required to explain a given variant as artifact, practitioners often need the converse: given an assumed platform error rate, how many reads must support a variant for it to be considered trustworthy? Table 2 inverts the analysis to answer this directly.

Table 2: Minimum variant read count M required for significance at assumed per-position error rates ($K = 1$, $L = 700$). Upper panel: conservative model ($q = p$). Lower panel: uniform model ($q = p/3$).

	$p = 1\%$			$p = 3\%$			$p = 5\%$		
N	0.05	10^{-3}	10^{-5}	0.05	10^{-3}	10^{-5}	0.05	10^{-3}	10^{-5}
Conservative ($q = p$)									
10	4	5	$6^\dagger$	5	$6^\dagger$	$7^\dagger$	5	$7^\dagger$	$8^\dagger$
20	4	5	7	6	7	9	7	9	10
50	6	7	9	9	11	13	11	13	15
100	7	9	11	12	15	17	16	19	22
200	10	12	15	18	21	24	25	28	32
500	16	19	22	32	37	41	46	52	57
1000	25	29	33	53	59	65	79	86	93
Uniform ($q = p/3$)									
10	3	4	5	4	5	$6^\dagger$	4	5	$6^\dagger$
20	3	4	5	4	5	7	5	6	8
50	4	5	7	6	7	9	7	9	10
100	5	6	8	7	9	11	9	11	13
200	6	8	10	10	12	15	13	16	18
500	9	11	14	16	19	22	22	26	29
1000	13	16	19	25	29	33	35	40	44

$^\dagger M > N/2$: the required read support exceeds half the total reads, making detection of multiple variants impossible at this sample size.

At the standard operating point ($N = 1000$) under the conservative model, a variant needs $M \geq 93$ reads to be significant at the run-corrected $\alpha = 10^{-5}$ when the assumed error rate is 5%, but only $M \geq 33$ at 1% assumed error. Under the uniform model, these thresholds drop to 44 and 19 respectively. At moderate coverage ($N = 100$), even the most stringent combination (5% error, $\alpha = 10^{-5}$, conservative) requires only $M \geq 22$, representing 22% of total reads.

3.5 Effect of multi-position evidence on minimum read support

Table 3 shows how the minimum M for significance changes when the variant is distinguished at $K > 1$ positions. Under the joint model (Section 2.5), requiring the same reads to error at all K positions dramatically reduces the evidence needed.

At $K = 2$, the minimum M drops to 3–6 reads across the full range of N; at $K = 3$, it is 3–4 reads regardless of sample size. Under the independence assumption, these thresholds are at or below the minimum count filters typically applied in practice (e.g., `--min-size 5`). However, as noted in Section 2.5, the $K > 1$ thresholds are sensitive to cross-position error correlation: at $K = 2$, $N = 1000$, the $M = 6$ threshold tolerates only a $\sim 2\times$ increase in joint error probability. For variants whose distinguishing positions are well-separated in the amplicon, the independence assumption is reasonable and these low thresholds apply. For nearby positions where error correlation is plausible, the $K = 1$ thresholds from Table 2 provide the conservative bound.

Table 3: Minimum variant read count M at $K = 1$, 2, and 3 variant positions (uniform model $q = p/3$, $\alpha = 10^{-5}$, $p = 1.5\%$, $L = 700$).

N	$K = 1$	$K = 2$	$K = 3$
10	5	3	3
20	6	3	3
50	7	4	3
100	9	4	3
200	11	4	3
500	16	5	3
1000	23	6	4

4 Discussion

4.1 Interpreting the critical error rate

The critical error rate p^* represents the per-position error rate at which a given variant pattern becomes statistically plausible as an artifact. When the conservative model is used ($q = p$), p^* can be interpreted as the rate of maximally biased error at a single position—the worst case where all errors produce the same incorrect base. When the uniform model is used ($q = p/3$), p^* represents the overall per-position error rate under the assumption that errors are unbiased across alternative bases.

For Oxford Nanopore reads, per-position substitution error rates typically range from 1–5% for simplex reads (Q13–Q20) and below 1% for duplex or consensus-corrected reads. Modern R10.4.1 chemistry achieves raw read accuracy of 98.7–99.4% (Q20+), corresponding to per-position error rates of 0.6–1.3% [8], making the margins above even more comfortable for well-supported variants. Variants with p^* substantially above these ranges are robust; variants with p^* near or below these ranges warrant caution.

4.2 Conservative framing

Several deliberate conservatisms ensure that critical error rates represent lower bounds on variant significance:

1. **Denominator choice.** Using the total input read count N rather than the cluster size assumes the worst case: the clustering algorithm could have drawn any subset from the full pool. In practice, coarse clustering first separates clearly unrelated sequences (e.g., different loci or contaminant reads), reducing the effective pool. However, finer-grained cluster boundaries are themselves determined by the variant signal under evaluation, so using cluster size as the denominator would be circular. The analysis uses total input reads to avoid this dependency.

2. **Maximally biased substitution.** The conservative model ($q = p$) assumes all errors at a position produce the same alternative base. If errors were spread across multiple alternatives ($q = p/3$), coherent error patterns would be less likely.

3. **Bonferroni correction.** Treating all L positions as independent tests is conservative; nearby positions in amplicon sequences are not truly independent, so the effective number of tests is smaller than L.

4. **Minimum K.** Using $K = 1$ represents the weakest case; most variants are distinguished at multiple positions. Because the same reads must error at all K positions independently, significance increases dramatically with K (Section 2.5).

4.3 Homopolymer normalization

The variant detection algorithm includes homopolymer normalization, which subtracts homopolymer length variation from the base composition at each position before evaluating variant thresholds. This removes the largest source of systematic, position-correlated nanopore error from consideration—approximately 47% of nanopore errors are homopolymer-related, with deletions dominating over mismatches and insertions [2]. The null model presented here does not account for this additional filtering, making the analysis further conservative: the effective error rate after homopolymer normalization is lower than the raw per-position rate.

This approach is conservative but lossy: it treats all homopolymer length differences as potential artifacts, even at short hp lengths (1–4 bases) where published accuracy exceeds 95% [6] and the error rate may be comparable to non-hp positions. A variant that differs from the consensus only by a single-base insertion within a dinucleotide repeat is discarded identically to one within a length-10 poly-G run, despite the former being far more likely to reflect real biology.

Extending the CER framework to evaluate hp-position variants statistically—rather than discarding them unconditionally—would require quantitative characterization of hp error rates by base and run length under current chemistry and basecaller versions. No published study provides this data for the R10.4.1 chemistry and Dorado basecallers used in our pipeline. With such data, the critical error rate analysis developed here could be applied to hp positions using empirical per-position error rates parameterized by hp context, enabling recovery of biological hp-length variants that the current blanket normalization discards.

4.4 Relationship to existing variant calling frameworks

Prior work on low-frequency variant detection has used binomial and Poisson-binomial models to test whether an observed allele count is consistent with sequencing error at an assumed rate. LoFreq [11] uses per-read Phred quality scores to construct a position-specific Poisson-binomial null distribution, applying Bonferroni correction across genomic positions. SNVer [9] uses an aggregate binomial model with Simes' procedure for multiplicity control. Our framework inverts the question: rather than testing significance at a fixed error rate, we derive the critical error rate p^* at which a variant becomes plausible as artifact. This inversion is both more conservative—it does not depend on the accuracy of per-read quality score calibration, which can be systematically overestimated for nanopore data [2]—and more interpretable for practitioners, who can directly compare p^* against known platform error rates.

4.5 Environmental and mixed specimens

The analysis assumes that N reads originate from sequencing a single organism, as is typical in specimen-based amplicon barcoding. For environmental DNA samples or specimens containing multiple organisms, the interpretation of N requires modification. When the read pool contains sequences from multiple distinct organisms, the relevant N for evaluating intra-organism variants is the number of reads from the target organism, not the total. The upstream clustering step partially addresses this by separating reads from different organisms into distinct clusters, but the formal guarantee applies within the assumption that the initial cluster contains reads from a single source.

4.6 Limitations

The null model assumes independence of errors across reads at each position. Nanopore errors exhibit well-documented sequence-context dependencies—certain k-mer contexts systematically produce higher error rates and biased substitution patterns [2]. The homopolymer normalization step addresses the largest class of such correlation, and the conservative model ($q = p$) accounts for substitution directionality by assuming all errors at a position produce the same alternative base. However, neither addresses the possibility that context-dependent effects elevate the per-position error rate at specific sites substantially above the platform average. The critical error rate framework is designed to accommodate this uncertainty: rather than assuming a fixed error rate, it reports p^* and leaves the reader to judge whether that rate is plausible given their platform, chemistry, and sequence context. A variant with $p^* = 6.8\%$ is robust if the worst-case positional error rate is 2%, but not if context effects at that position could plausibly reach 7%. As nanopore chemistry continues to improve, both mean error rates and context-dependent profiles change, making fixed thresholds unreliable; p^* remains interpretable against whatever platform characterization is current.

This framework addresses bioinformatic artifacts—sequences that never physically existed in the sequenced library. PCR errors introduced early in amplification produce sequences that physically exist in the library, and are therefore out of scope by design: passing the critical error rate test establishes physical presence in the library, not biological reality. For taxonomic identification, the primary mitigation against PCR artifacts and other non-biological sequences is robust reference databases where variants have been replicated across independent specimens.

5 Conclusion

The critical error rate framework provides a transparent basis for evaluating confidence in detected variants. Under typical operating conditions ($N = 1000$, $M = 100$), critical rates of 5.5–6.8% (conservative) or 16.6–20.3% (uniform) are well above realistic nanopore error, even at run-corrected significance levels. At lower coverage ($N = 100$, $M = 10$), the conservative model yields $p^* =$ 0.8–2.2%, approaching the boundary of distinguishability for lower-quality reads. The framework distinguishes physical library presence from bioinformatic hallucination; biological significance of detected variants requires downstream validation against reference databases. Table 2 provides direct lookup of the minimum read support needed at assumed error rates of 1–5%, enabling practitioners to assess variant confidence without computation. An important next step is extending this framework to homopolymer positions, where blanket normalization currently discards variants that may be biological; this requires empirical characterization of hp error rates by base and run length under current sequencing chemistry.

Acknowledgments

Substantial portions of the text were drafted with the assistance of Claude (Anthropic, Opus 4.6). The core ideas, analytical framing, and mathematical approach were provided by the author; the AI contributed to exposition, LaTeX formatting, and iterative drafting under human direction. The author takes full responsibility for the content.

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